

sub-gigahertz all the way to millimetre-wave, depending on the kind of usage profile that is needed for a specific use-case.

Its deployment can be diverse—it can be in the form of a micro network managed by an operator to a 5G network managed by a big industrial house. Such industrial houses can also set up their own 5G network. 3GPP gives the flexibility of having a private 5G network.

Examples of service offerings enabled by a private 5G network are:

- Ultra-Reliable Low-Latency Communication (URLLC), which provides highly reliable data throughputs
- Time-sensitive network, which is essentially wireless Ethernet that can help replace wired installations in industries
- Dedicated licensed or unlicensed spectrums, which enable deployment of 5G in private networks

Standardisation bodies, operators and infra-providers are working on technologies that will truly enable an extremely industrialised, mechanised and automatic way of running an industry. So, going forward, with the foundation on LTE IoT, an evolution will happen in the next decade. 5G and all other enhancements that are happening in 3GPP will drive these requirements.

Key industrial IoT functionalities targeted for 3GPP Release 16

In the past, 3GPP has been mainly focussed on specifications and operators. Now, the organisation is taking inputs from the industry. Work is going on in 3GPP Release 16 and beyond while taking inputs from industrial giants, who are now part of the standardisation body.

URLLC will provide low latency, highly reliable information, which the industry needs because it works in real time. And time-sensitive networks or wireless Ethernet, as it is also called, will provide high speed throughputs. Sensors sending huge amounts of data, cameras or head-mounted displays sending information back to the cloud will require such time-sensitive networks. Also, 5G will offer a flexible spectrum, all the way from sub-gigahertz to millimetre-wave that lasts for flexible deployments.

A combination of security, connectivity and compute is important for ensuring that we have a truly 5G-enabled IIoT ecosystem or paradigm in the coming decade.

5G deployments in India

Is India ready for 5G? It is only a matter of when and not if.

The government is firmly pushing it as a policy agenda to have 5G rolled out. With multiple spectrum options, a stable standardisation regime that is happening on 5G NR is being devised by 3GPP. Everything is converging to enable a truly 5G-based IIoT ecosystem, not only in India but across the world, in the not too distant future.

Seven to eight years ago when 4G was rolled out, it was quite expensive—now, it is very cheap. As technology gets mature, the ecosystem gets confidence and costs go down. It always happens with technology, and it will happen with 5G, too. **EFY**

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